

FIELDNODE · REQUIREMENTS

FieldNode requirements

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DOCUMENT CONTROL

Document ID	FND-REQ-001	Version	0.3
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/fieldnode		
Source	docs/03-requirements.md @ 536034e		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from FND-CAL-001; R11 and R16 restated to match FND-DDR-001 (D5 and the FieldNode core costing)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status against each
0.1	2026-09-25	Amish Chadha	Initial scaffold

FieldNode requirements

Eleven of the eighteen requirements are met on paper or by design, five are at risk, one is not met and one can only be verified by a timed installation (FND-CAL-001 v0.1, Table 4). The miss is R3: the enclosure runs hotter than 60 °C in full sun at 45 °C ambient, which also blocks cell charging in hot weather. Targets are unchanged from v0.2 except R11 and R16, which are restated to match FND-DDR-001; the design choices behind them are adopted for TRL 3 work pending Amish's review. The status column gives the standing from FND-CAL-001; "Met on paper" means shown by calculation, not by test.

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (FND-CAL-001)
R1	Weather protection	Enclosure IP65 or better with membrane vent; all penetrations sealed glands or rated connectors	Enclosure rating; later spray test	Met by design: IP65 box, ePTFE vent, two IP68 glands, capped IP67 ports and bulkhead
R2	Operating temperature	-20 to +45 °C ambient; electronics rated -20 to +70 °C inside the enclosure	Datasheets; thermal calculation	At risk: 73.3 °C inside in the dusty worst case; no charging on clear days colder than about -13 °C
R3	Interior temperature in sun	60 °C or less inside the enclosure at 45 °C ambient, full sun, still air	Thermal calculation	Not met: 58.7 °C clean and 66.2 °C dusty on the hot design day, 63.3 °C with the sun in its worst position; 48.5 °C with the proposed sun shield
R4	Safe charging window	Cell charging blocked below 0 °C and above 45 °C by an NTC on the cell	Charger design review	Met by design
R5	Energy neutral in the worst month	Daily harvest exceeds full-load demand at 2 peak sun hours	Energy budget	At risk: 7.75 Wh stored against 2.67 Wh drawn in the worst month, but only 0.8 Wh stored on a hot clear day because the cell is above 45 °C; 6 V class panel voltage marginal when hot
R6	Autonomy without sun	5 days or more at full sensor allowance, starting from 80 % charge usable	Energy budget	At risk: 5.75 days at 100 mW, exactly 5.00 days at 115 mW; 4.03 days at -20 °C and 4.60 days at end of life
R7	Sensor power allowance	100 mW average or more available to the sensor ports	Energy budget	Met on paper: 115.0 mW is the ceiling for 5 days; 100 mW design value
R8	Radio link	LoRaWAN uplink to a gateway 2 km away in suburban terrain at SF9 or faster	Link budget; later field test	Met on paper: 18.6 dB margin at 2 km to a 30 m gateway
R9	Airtime within fair use	30 s uplink airtime per day or less at the default interval	Airtime calculation	At risk: 23.7 s at SF9; not met at SF10 (43.5 s) to SF12 unless firmware lengthens the interval
R10	Store and forward	30 days or more of readings kept on the node when the link is down	Memory calculation	Met on paper: 30 days in 90 kB of 16 MB
R11	Standard sensor interface	Two sealed M12 5-pin ports, each carrying I2C, UART or RS-485, one analog input and one switched rail selectable at 3.3, 5 or 12 V; a gland for fixed-cable sensors	Pinout review with adopting projects	Met on paper; pinout open (FND-DDR-001, O2)
R12	Mounting and install	Fits 40 to 60 mm poles and flat walls; one person installs in 15 min or less with hand tools	Design review; later timed install	Fit met by design (seats 40 to 71 mm); time estimated at 15 min, not verifiable at TRL 3
R13	Wind	Survives 35 m/s (126 km/h) gusts on the panel without loosening	Load calculation	Met on paper: clamp pull 59 N against 2,000 N; band preload assumed

ID	Requirement	Target	Verification (TRL 3 or later)	Status (FND-CAL-001)
R14	Mass	2.5 kg or less including panel and mounts	Massing model, then weighing	At risk: 2.41 kg; 2.56 kg with the proposed shield
R15	Serviceable	Cell replaced in 10 min or less with a screwdriver; no soldering in the field	Design review	Met by design: about 7 min
R16	Cost	FieldNode core (enclosure, power, radio, panel and mounting) \$150 or less at quantity 1; sensors and gateway excluded	Priced BOM	Met on paper: \$126.00
R17	Open and independent	All design files under CERN-OHL-S-2.0 and MIT; works with any LoRaWAN network server, no closed cloud	Design review	Met by design
R18	Privacy pass-through	The core sends only what the sensor firmware passes to it; no images or audio leave the node	Firmware design review	Met by design; each project states its own rule

■ Requirements not met or at risk

- **R3 (interior temperature) not met.** The panel hood shades most of the top but none of the front, so low and side sun heat the box. Proposed, awaiting Amish: a ventilated white sun shield (about \$8, 0.15 kg), which keeps the inside to about 50 °C.
- **R5 and R2 (hot and cold charging) at risk.** Without the shield a run of hot, clear days drains a full cell in about 8 days, because the charge lockout holds above 45 °C. In the cold the cell charges only on clear days warmer than about -13 °C. A 6 V class panel may fall below the charger's minimum input when hot; a 9 V class panel is proposed.
- **R6 (autonomy) at risk.** 115 mW is exactly the 5-day ceiling; publishing 100 mW as the allowance gives 15 % margin, but cold or an aged cell still falls short.
- **R9 (airtime) at risk.** Nodes at SF10 or slower must report every 22 min or less often.
- **R14 (mass) at risk.** 2.41 kg, 0.09 kg under the limit; the shield would exceed it.

■ Assumptions

- Assumptions for every status above are stated in FND-CAL-001, Table 1. The main ones: worst month 2 peak sun hours on the tilted panel; panel, dust and angle losses 20 %; MPPT charger 85 %; cell charge 95 %; rail converters 90 %; LiFePO4 cell 3.2 V, 6 Ah (about 19 Wh), 80 % usable.
- Uplink payload 20 bytes plus 13 bytes of LoRaWAN overhead, 125 kHz bandwidth, coding rate 4/5, 8-symbol preamble.
- Core consumption from SX1262 datasheet figures ([Semtech SX1261/2 datasheet](#)): 4.0 mWh a day.